

Team Name

TrackMind AI

Smart India Hackathon • PS 26027



Team Members Names

- Sk.Kalesha Vali (Team Lead)
 - K.Vaishika • G.Navya Sri • K.Nikitha
 - M.Sravani • M.Manikanta
- ✉ n220895@gmail.com | ☎ 8498898017

Title AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways (Smart India Hackathon PS 26027)

- **Problem:** Three departments (Track, Signals, Traction) request isolated blocks, causing 8+ hours of daily corridor downtime & train delays.
- **Engine:** TrackMind AI unifies TMS, SMMS, TDMS, timetables & freight forecasts into a single AI optimization pipeline.
- **Optimization:** Random Forest breakdown risk ML + Google OR-Tools CP-SAT solver schedule concurrent Joint Shadow Blocks.
- **Key Impact:** Zero train collisions, 4+ hours of daily downtime saved, 66% fewer line stoppages, and multilingual voice AI governance.

ABSTRACT

📌 Key Result: 4+ Hours Saved Daily | 100% Timetable Protection

Challenges / Limitations

📌 **Uncoordinated Line Stoppages**

Track, Signals, and OHE request isolated blocks on the same section, causing 3 separate shutdowns totaling 8+ hours of daily downtime.

📌 **Multi-Departmental Data Silos**

Absence of a real-time unified database linking TMS, SMMS, and TDMS with live timetables and Control Office freight forecasts.

📌 **Complex Math Constraints**

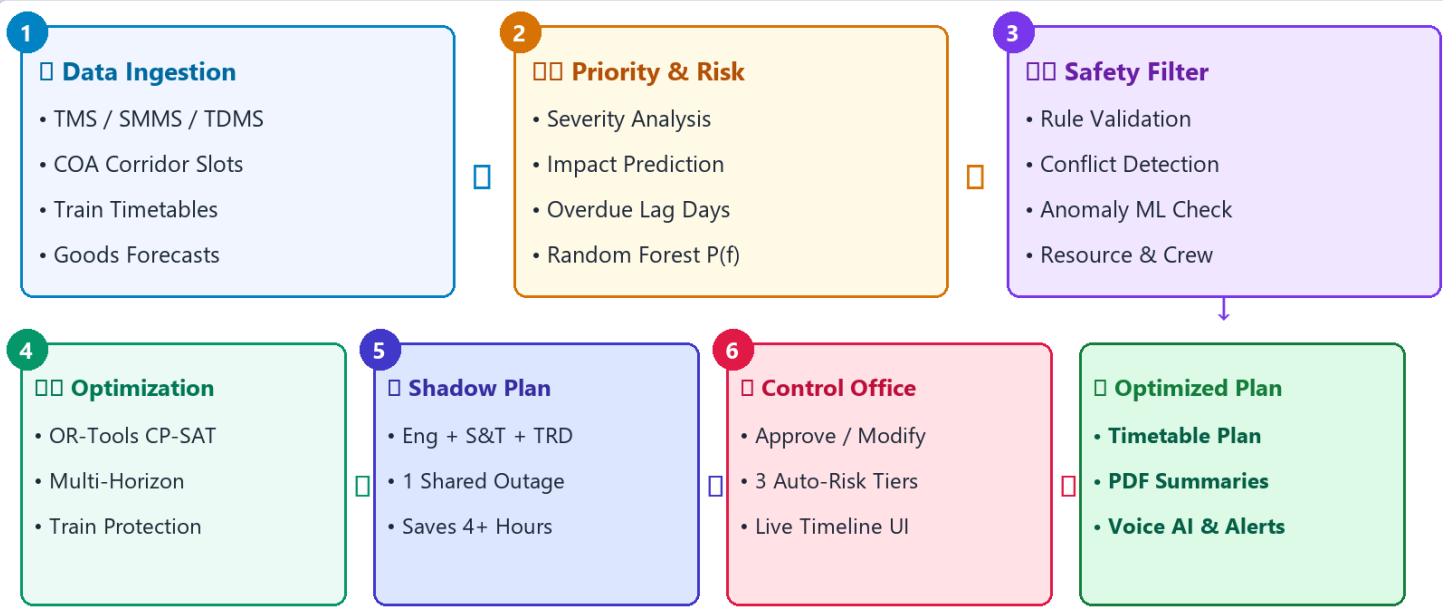
Balancing overdue safety defect SLAs, crew availability limits, and non-overlapping passenger train timetables is manually intractable.

📌 **Scope & API Limits**

Requires standardized digital API feeds from legacy railway software. Initial prototype uses SQLite; production transitions to Turso / PostgreSQL.

📌 SLA Governance: Monitored by 13 Autonomous Agents with 3 Risk Auto-Approval Tiers.

Workflow



📌 13 Autonomous Agents manage SLAs, crew rosters & auto-approvals.

1. Joint Shadow Block Clustering

Bundles Track (TMS), Signal (SMMS) & OHE (TDMS) into 1 shared outage—saving 4+ hrs daily!

2. Google OR-Tools CP-SAT Solver

Math constraint optimization for 7-day Weekly & 30-day Monthly zero-collision schedules.

3. Hybrid ML (Random Forest + Isolation)

Predicts breakdown risk P(f) & detects duration anomalies across 40 rail sections.

4. Dynamic Math Priority Scoring

Formula = 40% Severity + 25% Overdue Days + 20% Train Impact + 15% ML Failure Risk.

5. 13 Autonomous Governance Agents

Coordinates SLA safety compliance, cost optimization, crew rosters & public advisories.

6. 3-Tier Risk Auto-Approval Engine

Tier 1 Auto-Approve, Tier 2 Auto+Log, Tier 3 Critical Escalation for Controller Review.

7. Live Gantt & Plotly Timeline UI

Real-time zoomable corridor timeline visualization with live status & asset health logs.

8. Multilingual Voice AI (Groq LLM)

Llama-3.3-70B model with Web Speech API for voice in English, Hindi & Telugu.

📌 LIVE INTERACTIVE PROTOTYPE DEPLOYMENT LINK
<https://railway-block-planning.streamlit.app>

Credentials: admin1 / admin123